

# LECTURE 4: DERIVATIVES

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# PLAN

## Derivatives

1. Definition
2. Calculation
3. Higher Order Derivatives

## Optimization

1. Increasing/decreasing
2. Convex/concave
3. Local minima/maxima

## L'Hôpital's Rule



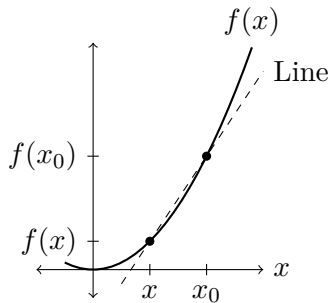
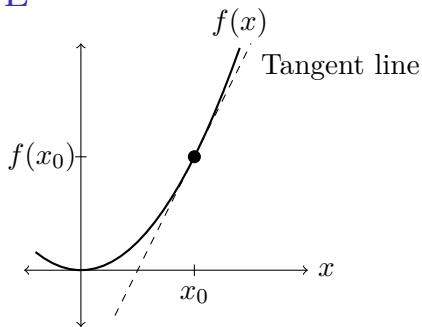
# TANGENT LINE

Consider the function  $f(x) = x^2$ .

We want to calculate the slope of the line that “touches” it at  $x_0 = 1$ .

We can approximate the tangent by lines that pass through  $(x_0, f(x_0))$  and  $(x, f(x))$  with  $x \rightarrow x_0$ .

The slope is  $\frac{f(x) - f(x_0)}{x - x_0}$ .



# DERIVATIVES

It makes sense to define the **derivative** of  $f$  at  $x_0$  as the limit

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

if it exists. If it exists and is not infinite, we say that  $f$  is **differentiable** at  $x_0$ .

Note that  $x \rightarrow x_0$  is the same as  $h \equiv x - x_0 \rightarrow 0$ , and  $x = x_0 + (x - x_0) = x_0 + h$ , so we can write

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

## EXAMPLE

If  $f(x) = ax + b$  we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{a(x+h) + b - (ax+b)}{h} = \lim_{h \rightarrow 0} \frac{ah}{h} \\ &= \lim_{h \rightarrow 0} a = a. \end{aligned}$$

If  $f(x) = x^2$  then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x. \end{aligned}$$

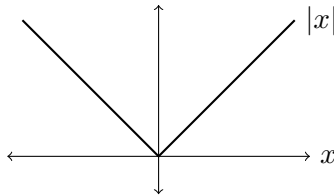
# DIFFERENTIABLE $\Rightarrow$ CONTINUOUS

We should think of the derivative as the rate of change of  $f(x)$  locally.

Notice that if  $f'(x_0)$  exists then  $f$  has to be continuous at  $x_0$ , because

$$\begin{aligned}\lim_{x \rightarrow x_0} f(x) &= \lim_{x \rightarrow x_0} (f(x) - f(x_0)) + f(x_0) \\ &= \lim_{x \rightarrow x_0} \left[ \frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] + f(x_0) \\ &= f'(x) \cdot 0 + f(x_0) = f(x_0).\end{aligned}$$

But not every continuous function is differentiable. For example,  $f(x) = |x|$  is not differentiable at 0.



# CALCULATING DERIVATIVES

If  $f$  and  $g$  are differentiable at  $x$ ,  $(f \pm g)'(x) = f'(x) \pm g'(x)$ . Why?

$$\begin{aligned}(f \pm g)'(x) &= \lim_{h \rightarrow 0} \frac{(f(x+h) \pm g(x+h)) - (f(x) \pm g(x))}{h} \\ &= \lim_{h \rightarrow 0} \left( \frac{f(x+h) - f(x)}{h} \pm \frac{g(x+h) - g(x)}{h} \right) = f'(x) \pm g'(x).\end{aligned}$$

# CALCULATING DERIVATIVES

If  $f$  and  $g$  are differentiable at  $x$ ,  $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$ :

$$\begin{aligned}(fg)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(f(x+h) - f(x))g(x+h) - f(x)(g(x+h) - g(x))}{h} \\&= \lim_{h \rightarrow 0} \left[ \frac{f(x+h) - f(x)}{h} g(x+h) + f(x) \frac{g(x+h) - g(x)}{h} \right] \\&= f'(x)g(x) + f(x)g'(x).\end{aligned}$$

**Example.**  $(x^3)' = (x \cdot x^2)' = x' \cdot x^2 + x \cdot (x^2)' = x^2 + x \cdot (2x) = 3x^2$ .

In general,  $(x^n)' = nx^{n-1}$ .

# EXERCISE

## QUESTION 1

Calculate the derivative of the following functions:

1.  $f(t) = 14t - 7$

2.  $f(y) = y^3 + 3y^2 - 12$

3.  $f(x) = (x^2 + 1)(x^3 - 1)$

4.  $f(x) = x^6 + 5x^5 - 2x^2 + 8$

# CHAIN RULE

If  $g$  is differentiable at  $x$  and  $f$  is differentiable at  $g(x)$ , define  $(f \circ g)(x) = f(g(x))$ , and

$$\begin{aligned}(f \circ g)'(x) &= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{h} \\&= \lim_{h \rightarrow 0} \left[ \frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \frac{g(x+h) - g(x)}{h} \right] \\&= f'(g(x))g'(x).\end{aligned}$$

For example, if  $f(x) = \frac{1}{x}$  we have

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{x-(x+h)}{(x+h)x}}{h} = \lim_{h \rightarrow 0} \frac{-h}{(x+h)xh} = \lim_{h \rightarrow 0} \frac{-1}{(x+h)x} = -\frac{1}{x^2}.$$

Therefore,  $\left(\frac{1}{g(x)}\right)' = (f \circ g)'(x) = f'(g(x))g'(x) = -\frac{1}{g(x)^2}g'(x) = -\frac{g'(x)}{g(x)^2}.$

# CALCULATING DERIVATIVES

Notice that if  $a$  is constant then  $a' = 0$ .

Hence if  $a$  is constant then  $(af(x))' = a'f(x) + af'(x) = af'(x)$ .

And

$$\begin{aligned}\left(\frac{f(x)}{g(x)}\right)' &= \left(f(x) \cdot \frac{1}{g(x)}\right)' = f'(x) \frac{1}{g(x)} + f(x) \left(\frac{1}{g(x)}\right)' \\ &= \frac{f'(x)}{g(x)} + f(x) \frac{-g'(x)}{g(x)^2} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.\end{aligned}$$

.

# EXERCISES

## QUESTION 2

Calculate the derivative of the following functions:

1.  $f(x) = \frac{x}{x+1}$

2.  $f(x) = \frac{1}{x^2+1}.$

# EXP AND LOG

We have  $\exp'(x) = \exp(x)$ :

$$\begin{aligned}\exp'(x) &= \lim_{h \rightarrow 0} \frac{\exp(x+h) - \exp(x)}{h} = \lim_{h \rightarrow 0} \frac{\exp(x) \exp(h) - \exp(x)}{h} \\ &= \lim_{h \rightarrow 0} \exp(x) \frac{\exp(h) - 1}{h} = \exp(x) \underbrace{\lim_{h \rightarrow 0} \frac{\exp(h) - 1}{h}}_{=1} = \exp(x).\end{aligned}$$

The fact that  $\lim_{h \rightarrow 0} \frac{\exp(h)-1}{h} = 1$  follows from the definition of  $e$  but is not obvious.

$\log$  is differentiable, although I won't prove it now. Let's assume it's true.

Notice that  $\exp(\log(x)) = x$ .

Differentiating both sides we get  $\exp'(\log(x)) \log'(x) = 1$ .

Using  $\exp' = \exp$ , this is  $\exp(\log(x)) \log'(x) = 1$ , i.e.,  $\log'(x) = \frac{1}{x}$ .

# GENERAL POWER RULE

If  $k \in \mathbb{R}$  and  $f(x) = x^k$  for  $x > 0$  we have

$$\begin{aligned} f'(x) &= (x^k)' \\ &= (\exp(\log(x^k)))' \\ &= (\exp(k \log(x)))' \\ &= \exp'(k \log(x)) k \log'(x) \\ &= \exp(\log(x^k)) k x^{-1} \\ &= x^k k x^{-1} = k x^{k-1}. \end{aligned}$$

## QUESTION 3

Calculate the derivative of  $f(x) = a^x$  for every  $a > 0$ .

# EXERCISES

## QUESTION 4

Calculate the derivative of the following functions:

1.  $f(x) = 3x^{1/3}$

2.  $f(x) = \frac{1}{100}x^{25} - \frac{1}{10}x^{0.25}$

3.  $f(y) = 1 - 1/y^2$

4.  $f(x) = \ln(2\pi x^2)$

5.  $f(x) = e^{-x^2}$

# HIGHER ORDER DERIVATIVES

If  $f : U \rightarrow \mathbb{R}$  is **differentiable** (i.e., differentiable at every  $x \in U$ ) then  $f' : U \rightarrow \mathbb{R}$  is another function, and we can ask if it is itself differentiable.

**Example.** Take  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0, \\ -x^2 & \text{if } x < 0. \end{cases}$$

Notice that it is differentiable. This is only non-obvious at 0. We have

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h^2}{h} = \lim_{h \rightarrow 0^+} h = 0, \quad \text{and} \\ \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^-} \frac{-h^2}{h} = \lim_{h \rightarrow 0^-} -h = 0, \end{aligned}$$

$$\text{so } f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 0.$$

# HIGHER ORDER DERIVATIVES

We obtain

$$f'(x) = \begin{cases} 2x & \text{if } x \geq 0, \\ -2x & \text{if } x < 0. \end{cases}$$

In other words,  $f'(x) = 2|x|$ . This is not differentiable at 0. So  $f'$  is not differentiable.

But if  $f'$  is differentiable at  $x$  we can define the **second derivative**  $f''(x)$  of  $f$  as the derivative of  $f'$  at  $x$ .

## QUESTION 5

If  $f(x) = x^3$  calculate  $f''(x)$  for all  $x \in \mathbb{R}$ .

# OPTIMIZATION

If  $f : D \rightarrow \mathbb{R}$  is maximized or minimized at  $x$  in the interior of  $D$  and  $f'(x)$  exists then  $f'(x) = 0$ .

The reason is that if  $f'(x) > 0$  then when  $h > 0$  is sufficiently small we have  $\frac{1}{h}(f(x+h) - f(x)) \approx f'(x) > 0$ , so  $f(x+h) > f(x)$ , and  $f(x)$  is not the maximum value of  $f$ . If  $f'(x) < 0$  then when  $h < 0$  is small we have  $\frac{1}{h}(f(x+h) - f(x)) \approx f'(x) < 0$ , so multiplying by  $h$  we get  $f(x+h) - f(x) > 0$ , and again  $f(x)$  is not the maximum.

If  $f : (a, b) \rightarrow \mathbb{R}$  is differentiable and  $f' > 0$  then  $f$  is increasing. (Intuition:  $f'(x) > 0$  means that  $f$  has a “positive slope” at  $x$ .)

If  $f' < 0$  then  $f$  is decreasing.

This is very useful! Example: what is the maximum of  $f(x) = x - x^2$ ? We have  $f'(x) = 1 - 2x$ , so  $f' > 0$  if  $x < \frac{1}{2}$  and  $f' < 0$  if  $x > \frac{1}{2}$ . Hence  $f$  increases and then decreases. Therefore the max is at  $x = \frac{1}{2}$ .

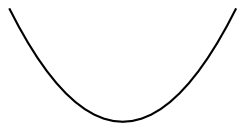
## EXAMPLE

### QUESTION 6

Find the minimum and maximum of  $f : [0, 1] \rightarrow \mathbb{R}$  given by  $f(x) = 2x^2 - x + 1$ .

# CONVEXITY

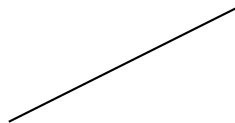
If  $f$  is twice differentiable then we say it's **convex** if  $f'' \geq 0$ , and **concave** if  $f'' \leq 0$ .



Convex



Concave



Both

If  $f$  is concave and  $f'(x) = 0$  then  $f$  is maximized at  $x$ .

If  $f$  is convex and  $f'(x) = 0$  then  $f$  is minimized at  $x$ .

If  $f''$  exists around  $x$ , is continuous at  $x$ , and  $f''(x) < 0$  then  $f'' < 0$  around  $x$ , so  $f$  is concave around  $x$ , and, so, if  $f'(x) = 0$  then  $f$  has a **local maximum** at  $x$ .

If  $f''$  exists around  $x$ , is continuous at  $x$ ,  $f''(x) > 0$ , and  $f'(x) = 0$  then  $f$  has a **local minimum** at  $x$ .

## EXAMPLE

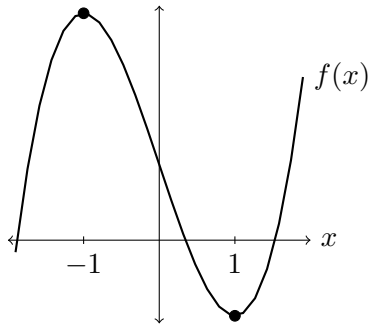
Take  $f(x) = x^3 - 3x + 1$ .

We have  $f'(x) = 3x^2 - 3$  and  $f''(x) = 6x$ .

$f'(x) = 0$  is  $3x^2 = 3$ , i.e.,  $x = -1$  or  $x = 1$ .

$f''(-1) = -6 < 0$ , so  $f$  has a local maximum at  $-1$ .

$f''(1) = 6 > 0$ , so  $f$  has a local minimum at  $1$ .

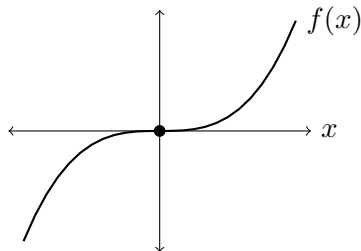


## WHAT IF $f''(x) = 0$ ?

Take  $f(x) = x^3$ .

We have  $f'(x) = 3x^2$  and  $f''(x) = 6x$ .

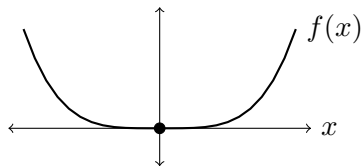
We have  $f'(0) = 0$  but  $f''(0) = 0$ , so it's not obvious if it's a local minimum or maximum. It's neither.



Take  $f(x) = x^4$ .

We have  $f'(x) = 4x^3$  and  $f''(x) = 12x^2 \geq 0$ , so it's convex.

We have  $f'(0) = 0$  but  $f''(0) = 0$ . But  $f$  is convex, so it must be minimized at 0.



# EXERCISE

## QUESTION 7

Find the minimum and maximum of the following functions:

1.  $f : [0, 1] \rightarrow \mathbb{R}, f(x) = x(1 - x).$
2.  $f : [0, 1] \rightarrow \mathbb{R}, f(x) = x - \sqrt{x}.$
3.  $f : [0, 6] \rightarrow \mathbb{R}, f(x) = x^3 - \frac{15}{2}x^2 + 12x + 8.$
4.  $f : [-4, 4] \rightarrow \mathbb{R}, f(x) = 3x^4 - 4x^3 - 36x^2.$

# L'HÔPITAL'S RULE

## THEOREM

Suppose  $\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x)$ , both are 0 or  $\pm\infty$ ,  $f, g$  are differentiable near  $x_0$ , and  $g' \neq 0$  near  $x_0$ . Then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)},$$

provided the latter limit exists.

## Examples.

1.  $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^3 - 8} = \lim_{x \rightarrow 2} \frac{2x}{3x^2} = \frac{1}{3}.$

2.  $\lim_{x \rightarrow +\infty} \frac{e^x}{x^2} = \lim_{x \rightarrow +\infty} \frac{e^x}{2x} = \lim_{x \rightarrow +\infty} \frac{e^x}{2} = +\infty.$

3.  $\lim_{x \rightarrow 0} x \log(x) = \lim_{x \rightarrow 0} \frac{\log(x)}{1/x} = \lim_{x \rightarrow 0} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0} (-x) = 0.$

# EXERCISE

## QUESTION 9

Calculate the following limits:

1.  $\lim_{x \rightarrow +\infty} \frac{\log(x)}{x}.$

2.  $\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1}.$

3.  $\lim_{x \rightarrow 0^+} x^x.$

## QUESTION 10

Find the minimum and maximum value of  $f : (0, 1] \rightarrow \mathbb{R}$  given by  $f(x) = x \log(x)$ .

